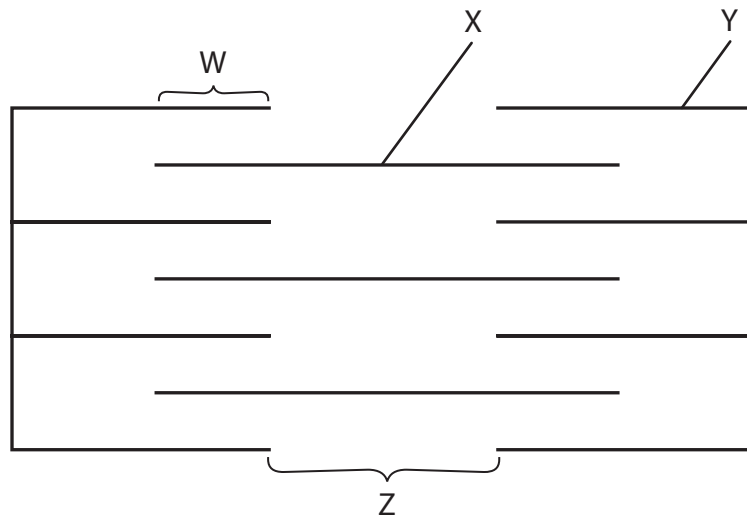


4 Skeletal muscles function to move the limbs of an organism.

(a) The diagram represents a sarcomere from skeletal muscle.



(i) Which protein forms the structure labelled X?

(1)

- ☐ **A** actin
- ☐ **B** myosin
- ☐ **C** tropomyosin
- ☐ **D** troponin

(ii) Which part becomes shorter when the muscle contracts?

(1)

- ☐ **A** W
- ☐ **B** X
- ☐ **C** Y
- ☐ **D** Z

(iii) How many of the following statements are correct when a sarcomere contracts?

- calcium ions bind to troponin
- calcium ions are released from the sarcoplasmic reticulum
- myosin changes shape

(1)

☐ **A** none

☐ **B** one

☐ **C** two

☐ **D** three

(iv) Which molecule contains an enzyme that hydrolyses ATP?

(1)

☐ **A** actin

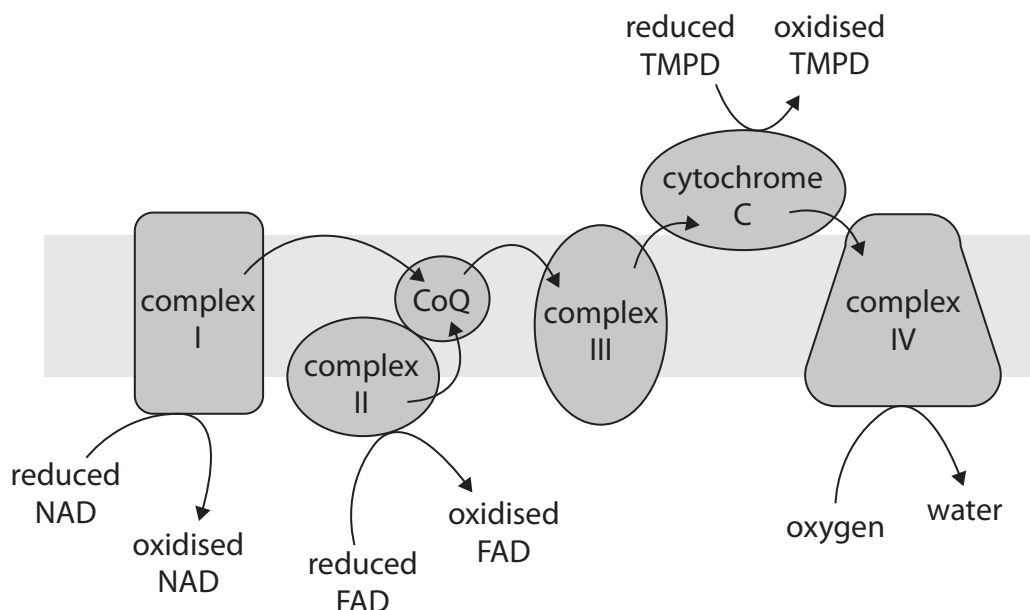
☐ **B** myosin

☐ **C** tropomyosin

☐ **D** troponin

*(b) Inherited mitochondrial diseases affect muscle function.

The diagram shows the electron transport chain in a mitochondrion.



In an investigation, samples of muscle tissue from healthy individuals and from four individuals with mitochondrial disease were collected.

The function of mitochondria, when these samples were treated with different substrates, was investigated.

Treating muscle tissue from healthy individuals with:

- pyruvate and malate increases the production of reduced NAD
- succinate increases the production of reduced FAD
- reduced TMPD increases the reduction of cytochrome c.

The oxygen consumption of the muscle tissue samples from healthy individuals and from four individuals with mitochondrial disease were recorded.

The results are shown in the table.

Sample taken from	Oxygen consumption in the presence of each treatment / $\text{pmol O}_2 \text{ min}^{-1} \text{ mg}^{-1}$ of muscle tissue		
	pyruvate and malate	succinate	reduced TMPD
Individual 1	3.0	3.5	13.0
Individual 2	18.0	10.0	60.0
Individual 3	6.5	14.0	51.0
Individual 4	7.5	6.0	40.0
Healthy individuals	21.0 ± 5.0	13.5 ± 4.5	63.0 ± 20.0

Comment on the defects in the electron transport chain in the individuals with mitochondrial disease.

This image shows a single sheet of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting or typing. There are no margins, text, or other markings on the paper.

(Total for Question 4 = 10 marks)